ASSET MANAGEMENT MODULE

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 · March 2026
Confidential · Internal Documentation

Audience: Development Team · Product Team · School Administrators · Finance Department

Related Modules:
Finance · Procurement · Store & Inventory · Maintenance · Staff Management · IT Management

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1. MODULE OVERVIEW

1.1 Purpose

The **Asset Management Module** manages all long-term physical assets owned by the school.

Assets include:

- Computers
- Vehicles
- Laboratory equipment
- Classroom furniture
- Sports equipment
- Buildings
- Library equipment
- IT infrastructure

The system ensures **accurate tracking, depreciation accounting, asset utilization, and lifecycle management**.

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2. CORE OBJECTIVES

The module must allow institutions to:

1. Maintain a **centralized asset registry**
2. Track **asset location and ownership**
3. Monitor **asset depreciation**
4. Manage **maintenance schedules**
5. Track **asset assignments**
6. Control **asset lifecycle**
7. Maintain **financial compliance**
8. Generate **asset reports and audits**

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3. SYSTEM ROLES & ACCESS CONTROL

Role	Permissions
Super Admin	Platform-wide asset oversight
School Admin	Asset configuration
Asset Manager	Manage asset records
Maintenance Staff	Update maintenance records
Finance Officer	Manage depreciation and financial value
Department Head	View and request assets
Auditor	Read-only access

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4. MODULE ARCHITECTURE

The Asset Management Module contains the following subsystems:

```
Asset Management
│
├── Asset Dashboard
├── Asset Registry
├── Asset Categories
├── Asset Assignments
├── Depreciation Management
├── Maintenance Management
├── Asset Transfers
├── Asset Disposal
├── Warranty & Service Tracking
├── Asset Audit & Verification
├── Asset Reports & Analytics
```

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5. ASSET DASHBOARD

5.1 Overview

The **Asset Dashboard** provides a summary of asset health and financial value.

Dashboard Metrics

- Total Assets
- Gross Asset Value
- Net Book Value
- Accumulated Depreciation
- Assets Under Maintenance
- Assets Due for Replacement
- Warranty Expiry Alerts

Example Dashboard

```
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Asset Management Dashboard
---------------------------------------

Total Assets: 1,245
Gross Value: $3,250,000
Net Book Value: $2,420,000

Assets Under Repair: 12
Assets Due for Replacement: 5
Warranty Expiring Soon: 8
```

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6. ASSET REGISTRY

6.1 Overview

The **Asset Registry** stores records for all institutional assets.

Each asset includes:

- Identification
- Category
- Purchase details
- Financial value
- Location
- Assigned owner

6.2 Asset Data Model

```
Asset
- asset_id
- asset_name
- category_id
- asset_tag
- serial_number
- purchase_date
- purchase_cost
- current_value
- location
- assigned_to
- warranty_expiry
- condition_status
```

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7. ASSET CATEGORIES

Assets are grouped into logical categories.

Default Categories

- IT Equipment
- Transport
- Laboratory Equipment
- Classroom Furniture
- Sports Equipment
- Buildings & Infrastructure

Example:

```
IT Equipment
│
├── Desktop Computers
├── Laptops
├── Printers
```

Schools can add **custom asset categories**.

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8. ASSET TAGGING SYSTEM

Each asset receives a **unique asset tag**.

Example:

```
SC-IT-000124
SC-LAB-000045
SC-VEH-000003
```

Tags may be implemented as:

- QR codes
- Barcodes
- RFID tags

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9. ASSET ASSIGNMENT MANAGEMENT

Assets can be assigned to:

- Staff members
- Departments
- Classrooms
- Labs
- Vehicles

Example Assignment

```
Asset: Laptop Dell XPS
Assigned To: IT Department
Responsible Staff: John Smith
Location: Computer Lab 2
```

Assignment Data Model

```
AssetAssignment
- assignment_id
- asset_id
- assigned_to
- department
- location
- assignment_date
- return_date
```

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10. DEPRECIATION MANAGEMENT

Tracks asset value over time.

Depreciation methods supported:

- Straight-line depreciation
- Declining balance
- Units of production

Depreciation Data Model

```
AssetDepreciation
- depreciation_id
- asset_id
- depreciation_method
- useful_life_years
- residual_value
- annual_depreciation
```

Example Calculation

```
Laptop Cost: $1,200
Useful Life: 4 years
```

```
Annual Depreciation:
$300 per year
```

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11. MAINTENANCE MANAGEMENT

Tracks asset servicing and repairs.

Maintenance Types

- Preventive maintenance
- Corrective maintenance
- Emergency repair

Maintenance Data Model

```
AssetMaintenance
- maintenance_id
- asset_id
- maintenance_type
- description
- technician
- maintenance_date
- cost
```

Example Record

```
Asset: School Bus
Maintenance Type: Service
Cost: $350
Date: 10 March 2026
```

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12. WARRANTY & SERVICE TRACKING

Tracks warranties and service contracts.

Example:

```
Laptop Dell XPS
Warranty Expiry: 10 Aug 2027
Service Provider: Dell Kenya
```

Alerts are generated before warranty expiration.

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13. ASSET TRANSFER MANAGEMENT

Assets can be transferred between locations.

Example:

```
Projector
From: Classroom A
To: Conference Room
```

Transfer Data Model

```
AssetTransfer
- transfer_id
- asset_id
- from_location
- to_location
- transfer_date
```

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14. ASSET DISPOSAL MANAGEMENT

Handles retirement of assets.

Reasons:

- Obsolete
- Damaged beyond repair
- Sold
- Donated

Disposal Data Model

```
AssetDisposal
- disposal_id
- asset_id
- disposal_type
- disposal_date
- disposal_value
```

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15. ASSET AUDIT & VERIFICATION

Supports **periodic asset audits**.

Audit methods:

- Manual inspection
- QR code scanning
- RFID verification

Example audit:

```
Asset: Printer HP 4100
Status: Verified
Location: Administration Office
```

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16. REPORTS & ANALYTICS

Financial Reports

- Asset valuation
- Depreciation reports
- Capital expenditure reports

Operational Reports

- Asset allocation by department
- Maintenance costs
- Asset utilization

Example Report

```
IT Equipment Report

Total Assets: 120
Total Value: $150,000
Assets Under Maintenance: 3
```

17. ALERTS & NOTIFICATIONS

The system generates alerts for:

- Maintenance due
- Warranty expiration
- Asset disposal recommendations
- Asset audit schedules

Notifications can be sent via:

- In-app alerts
- Email
- SMS

18. SYSTEM INTEGRATIONS

Module	Purpose
Finance	Asset valuation and depreciation
Procurement	Asset purchasing
Inventory	Asset components
Maintenance	Repair tracking
Staff	Asset assignment
IT Management	Device management

19. SECURITY & AUDIT TRAIL

All actions are logged.

Example:

```
AuditLog
- log_id
- user_id
- action
- asset_id
- timestamp
```

Example entry:

```
User: Admin
Action: Updated Asset Value
Asset: Laptop Dell XPS
Date: 16 March 2026
```

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20. BULK DATA IMPORT / EXPORT

Supports bulk asset operations.

Formats:

- CSV
- Excel
- API integrations

Use cases:

- Import assets during system setup
- Export reports for accounting

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21. MOBILE ASSET MANAGEMENT

Mobile support allows staff to:

- Scan asset QR codes
- Update asset condition
- Record maintenance

- Perform asset audits

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22. FUTURE ENHANCEMENTS

Advanced capabilities:

- AI asset lifecycle prediction
- Predictive maintenance
- IoT equipment monitoring
- Smart asset tagging (RFID)
- Automated replacement planning